

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 4 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|-----------------------------------|--|
| 1. Orthonormal vectors C | (A) The system $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ whose solution is the best $\hat{\mathbf{x}}$. |
| 2. Orthogonal matrix Q E | (B) The process that turns independent vectors into an orthonormal set (and builds $A = QR$). |
| 3. Projection matrix P H | (C) Perpendicular <i>unit</i> vectors: $q_i \cdot q_j = 0$ and each $q_i \cdot q_i = 1$. |
| 4. Normal equations A | (D) The error vector $\mathbf{e} = \mathbf{b} - \mathbf{p}$; it lies in $N(A^T)$, perpendicular to $C(A)$. |
| 5. Least squares G | (E) A square matrix with orthonormal columns, so $Q^T Q = I$ and $Q^T = Q^{-1}$. |
| 6. Residual (error) D | (F) All vectors perpendicular to a subspace V ; together with V it fills the whole space. |
| 7. Orthogonal complement F | (G) Choosing $\hat{\mathbf{x}}$ to make the error $\ \mathbf{b} - A\mathbf{x}\ ^2$ as small as possible. |
| 8. Gram-Schmidt B | (H) The matrix $P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}$ that sends $\mathbf{b}$ to its projection $\mathbf{p} = P\mathbf{b}$. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The least-squares solution makes $A\hat{\mathbf{x}}$ the point of $C(A)$ that is (A) farthest from $\mathbf{b}$ (B) closest to $\mathbf{b}$ ← (C) equal to $\mathbf{b}$ (D) zero
- For an orthogonal matrix Q , the product $Q^T Q$ equals (A) 0 (B) Q (C) I ← (D) Q^2
- The least-squares error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to (A) every column of A ← (B) $\hat{\mathbf{x}}$ (C) nothing (D) $\mathbf{b}$
- The normal equations for the best solution of $A\mathbf{x} = \mathbf{b}$ are (A) $A\mathbf{x} = \mathbf{b}$ (B) $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ ← (C) $AA^T \mathbf{x} = \mathbf{b}$ (D) $\hat{\mathbf{x}} = \mathbf{b}$
- If Q has orthonormal columns, the coordinates of $\mathbf{b}$ in that basis are found by (A) solving a big system (B) the dot products $q_i \cdot \mathbf{b}$ ← (C) inverting $A^T A$ (D) guessing
- A projection matrix P always satisfies (A) $P^2 = P$ ← (B) $P^2 = I$ (C) $P = 0$ (D) $P^T = -P$

Teacher Note

Item 1: the best $A\hat{\mathbf{x}} = \mathbf{p}$ is the projection, the closest point of $C(A)$ to $\mathbf{b}$. Item 3: “best” = error perpendicular to the subspace, i.e. to every column, which is exactly $A^T \mathbf{e} = \mathbf{0}$. Item 5: orthonormal columns make $\hat{\mathbf{x}} = Q^T \mathbf{b}$, coordinate $i = q_i \cdot \mathbf{b}$. Item 6: projecting an already-projected vector changes nothing, so $P\mathbf{p} = \mathbf{p}$ and $P^2 = P$.

Part C — Short Answer & Computation (35 pts)

Show your work.

- Line V through $\mathbf{a} = (1, 2, 2)$; test $\mathbf{w}, \mathbf{u}$ for $V^\perp$. **(a) Yes. $\mathbf{a} \cdot \mathbf{w} = (1)(2) + (2)(-1) + (2)(0) = 2 - 2 + 0 = 0$, so $\mathbf{w} \perp \mathbf{a}$ and $\mathbf{w} \in V^\perp$. (b) Yes. $\mathbf{a} \cdot \mathbf{u} = (1)(0) + (2)(2) + (2)(-2) = 0 + 4 - 4 = 0$, so $\mathbf{u} \in V^\perp$ as well. ($V^\perp$ is the whole plane $x + 2y + 2z = 0$.)**
- Project $\mathbf{b} = (3, 1)$ onto the line through $\mathbf{a} = (1, 1)$. $\hat{x} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} = \frac{3+1}{1+1} = \frac{4}{2} = 2$. Then $\mathbf{p} = 2\mathbf{a} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\mathbf{e} = \mathbf{b} - \mathbf{p} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. **Check: $\mathbf{a} \cdot \mathbf{e} = (1)(1) + (1)(-1) = 0$, so $\mathbf{e} \perp \mathbf{a}$. ✓**
- Normal equations to project $\mathbf{b} = (6, 0, 0)$ onto $C(A)$, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$. $A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^T \mathbf{b} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$. **Solving $\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$ gives $\hat{\mathbf{x}} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Then $\mathbf{p} = A\hat{\mathbf{x}} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{e} = \mathbf{b} - \mathbf{p} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$. **Check: $A^T \mathbf{e} = \begin{bmatrix} 1-2+1 \\ 0-2+2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}$. ✓****
- Best line $y = C + Dt$ for $(0, 3), (1, 3), (2, 5), (3, 9)$. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ 5 \\ 9 \end{bmatrix}$. Then $A^T A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$, $A^T \mathbf{b} = \begin{bmatrix} 20 \\ 40 \end{bmatrix}$, **giving $(C, D) = (2, 2)$: the line $y = 2 + 2t$. Fitted $\mathbf{p} = (2, 4, 6, 8)$, residuals $\mathbf{e} = \mathbf{b} - \mathbf{p} = (1, -1, -1, 1)$, and $\sum e_i = 1 - 1 - 1 + 1 = 0$.**
- Show $\mathbf{q}_1 = \frac{1}{5}(3, 4)$, $\mathbf{q}_2 = \frac{1}{5}(4, -3)$ are orthonormal; why $Q^T Q = I$. $\mathbf{q}_1 \cdot \mathbf{q}_1 = \frac{1}{25}(9 + 16) = \frac{25}{25} = 1$; $\mathbf{q}_2 \cdot \mathbf{q}_2 = \frac{1}{25}(16 + 9) = 1$ **(both unit length)**. $\mathbf{q}_1 \cdot \mathbf{q}_2 = \frac{1}{25}(12 - 12) = 0$ **(perpendicular)**. **Entry (i, j) of $Q^T Q$ is exactly $\mathbf{q}_i \cdot \mathbf{q}_j$, so $Q^T Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.**
- Write $\mathbf{b} = (5, 0)$ as $c_1 \mathbf{q}_1 + c_2 \mathbf{q}_2$ without solving a system. $c_1 = \mathbf{q}_1 \cdot \mathbf{b} = \frac{1}{5}(3 \cdot 5 + 4 \cdot 0) = \frac{15}{5} = 3$; $c_2 = \mathbf{q}_2 \cdot \mathbf{b} = \frac{1}{5}(4 \cdot 5 + (-3) \cdot 0) = \frac{20}{5} = 4$. **So $\mathbf{b} = 3\mathbf{q}_1 + 4\mathbf{q}_2$. (Dot products give the coordinates directly because the basis is orthonormal.)**
- Gram-Schmidt on $\mathbf{a}_1 = (3, 4)$, $\mathbf{a}_2 = (1, 0)$. **Normalize $\mathbf{a}_1$: $\|\mathbf{a}_1\| = \sqrt{9 + 16} = 5$, so $\mathbf{q}_1 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Remove the $\mathbf{q}_1$ -part of $\mathbf{a}_2$: $\mathbf{q}_1 \cdot \mathbf{a}_2 = \frac{3}{5}$, so $\mathbf{B} = \mathbf{a}_2 - \frac{3}{5}\mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \frac{3}{25}\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 16/25 \\ -12/25 \end{bmatrix}$. **Normalize: $\|\mathbf{B}\| = \frac{4}{5}$, so $\mathbf{q}_2 = \frac{5}{4}\mathbf{B} = \frac{1}{5}\begin{bmatrix} 4 \\ -3 \end{bmatrix}$. Thus $Q = \frac{1}{5}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.****

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- Fitted values $\mathbf{p} = A\hat{\mathbf{x}}$, residuals $\mathbf{e} = \mathbf{b} - \mathbf{p}$. **(a) The best $\mathbf{p}$ is the point of $C(A)$ closest to $\mathbf{b}$, and the closest point is the foot of the perpendicular: the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ must be perpendicular to the whole subspace $C(A)$. Perpendicular to $C(A)$ means perpendicular to every column of A , i.e. $(\text{column}) \cdot \mathbf{e} = 0$ for each column — and stacking those dot products is exactly $A^T \mathbf{e} = \mathbf{0}$. (If $\mathbf{e}$ had any component along $C(A)$ we could move $\mathbf{p}$ and shrink the error, so it would not be best.) (b) An all-ones column means one of the equations in $A^T \mathbf{e} = \mathbf{0}$ is $(1, 1, \dots, 1) \cdot \mathbf{e} = 0$, i.e. $\sum e_i = 0$. So the residuals must sum to zero: the amounts the line overshoots exactly balance the amounts it undershoots.**
- Q square with orthonormal columns. **(a) Write $\mathbf{b} = \sum c_i \mathbf{q}_i$ and dot both sides with $\mathbf{q}_j$: $\mathbf{q}_j \cdot \mathbf{b} = \sum c_i (\mathbf{q}_j \cdot \mathbf{q}_i)$. Because the columns are orthonormal, $\mathbf{q}_j \cdot \mathbf{q}_i = 0$ except $\mathbf{q}_j \cdot \mathbf{q}_j = 1$, so the whole sum collapses to c_j . Hence $c_j = \mathbf{q}_j \cdot \mathbf{b}$ and $\mathbf{b} = \sum (\mathbf{q}_i \cdot \mathbf{b}) \mathbf{q}_i$ — the dot products are the coordinates, with no system to solve. (b) The normal equations are $Q^T Q \hat{\mathbf{x}} = Q^T \mathbf{b}$. Since $Q^T Q = I$, the left side is just $\hat{\mathbf{x}}$, so $\hat{\mathbf{x}} = Q^T \mathbf{b}$: the expensive $A^T A$ solve disappears when the columns are orthonormal.**

Scoring Part D (5 pts each). D1: 3 pts (a) closest point $\Rightarrow$ error $\perp C(A) \Rightarrow \perp$ every column $\Rightarrow A^T \mathbf{e} = \mathbf{0}$; 2 pts (b) the all-ones row of $A^T \mathbf{e} = \mathbf{0}$ is $\sum e_i = 0$ (over/undershoots cancel). D2: 3 pts (a) dot with q_j and use orthonormality to isolate $c_j = q_j \cdot \mathbf{b}$; 2 pts (b) $Q^T Q = I$ collapses the normal equations to $\hat{\mathbf{x}} = Q^T \mathbf{b}$. Accept equivalent wording throughout.